

A Systematic Comparative Study of Retrieval Methods for Parametric Construction Catalogs: From Lexical to Neural Approaches

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Abstract

Construction projects increasingly rely on large parametric catalogs containing tens of thousands of near-identical items, where accurate retrieval is critical for cost estimation and procurement. This paper presents a systematic comparison of lexical, neural, and hybrid information retrieval methods for matching short work descriptions against a large-scale parametric catalog of Spanish railway infrastructure. The catalog’s hierarchical structure enables a dual evaluation at item level (exact variant) and parent level (family). Across 100 configurations evaluated on approximately 47,500 catalog items, a tuned BM25 with domain-specific parameter phrases achieves 97.4 (BGE-M3-ColBERT, 44.8) more than 8 points. Neural embeddings remain competitive at the parent level (up to 99.7) numeric parameters. Well-tuned lexical retrieval thus provides a strong, interpretable baseline for parametric construction catalog search.

Keywords: Information retrieval Construction catalog BM25 Dense retrieval Parametric catalog Benchmark dataset

Glossary

Acc@1 Accuracy at rank 1 — the proportion of queries for which the correct item appears in the top position.

ADIF Spanish acronym for *Administrador de Infraestructuras Ferroviarias*, the public entity responsible for railway infrastructure in Spain.

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27 **BC3 Format** Base de Costes de la Construcción, version 3 — the standard
 28 data exchange format for Spanish construction cost databases, encoding
 29 hierarchical cost structures with parametric templates and work item
 30 descriptions.

31 **BM25** A probabilistic ranking function that scores document relevance
 32 based on term frequency and document length normalization.

33 **Dense Retrieval** A neural retrieval approach that represents queries and
 34 documents as dense embeddings, enabling semantic similarity search
 35 via vector spaces.

36 **E5 Model** A sentence-transformer-based embedding model trained for
 37 general-purpose retrieval and semantic similarity tasks.

38 **Hard-Negative Retrieval** A retrieval scenario where the correct item
 39 must be distinguished from highly similar candidates differing only in
 40 one or more critical attributes, such as a numeric specification.

41 **Hybrid Retrieval** A method that combines lexical and neural approaches,
 42 often through score fusion or reranking, to balance precision and se-
 43 mantic coverage.

44 **Item-Level / Parent-Level Evaluation** A dual evaluation protocol as-
 45 sessing retrieval at two granularities: item-level requires retrieving the
 46 exact parametric variant, while parent-level credits any item from the
 47 same parametric family.

48 **Language Model (LM)** A lexical retrieval method that estimates the
 49 probability of generating a query from a document’s language distribu-
 50 tion.

51 **MRR** Mean Reciprocal Rank — the average reciprocal of the rank position
 52 of the first relevant result across all queries.

53 **nDCG** Normalized Discounted Cumulative Gain — a ranking metric that
 54 accounts for graded relevance and result position.

55 **Parametric Catalog** A structured database where construction items are
 56 described by sets of parameters. Each unique combination of parameter
 57 values generates a distinct catalog item.

Pseudo-Relevance Feedback (PRF) A technique that expands the query using top-ranked documents from an initial retrieval round to improve recall.

1. Introduction

In modern construction project management, accurately retrieving the correct work item from standardized catalogs is critical for budget control, procurement and stakeholder communication [1, 2]. Large infrastructure projects increasingly rely on parametric construction catalogs containing tens of thousands of highly similar items, where small lexical or numeric differences correspond to distinct technical specifications and unit costs. In this context, catalog retrieval is a precision-critical task rather than a general semantic search problem.

The challenge is compounded by terminology variability across stakeholders. Field workers, designers, and cost estimators often use different terms for identical concepts, creating a vocabulary mismatch between queries and catalog entries [3, 4, 5, 6]. Moreover, construction descriptions frequently contain critical numeric parameters—dimensions, quantities, specifications—where even minor discrepancies can lead to costly errors in project estimation [2]. These domain-specific characteristics demand retrieval methods that balance lexical precision with semantic understanding while maintaining sensitivity to numeric values.

These terminology mismatches are particularly evident in two operational contexts. First, during cost estimation for design-bid-build tenders, estimators must map project specifications—often drafted with company-specific or region-specific wording—to standardized catalog entries. A project item description related to pipes must be matched to the precise catalog variant that accounts for terrain type, pipe diameter, and work-shift conditions, yet the source document may omit or paraphrase critical parameters. Second, during project execution, field engineers produce daily progress reports using abbreviated or colloquial item descriptions that must be reconciled with contractual line items for billing and compliance. A site note should map unambiguously to a specific catalog entry, but the terse phrasing and implicit context challenge both manual lookup and automated systems. In both scenarios, failure to retrieve the correct item carries tangible consequences: cost overruns from misquoted unit prices, contractual disputes over scope interpretation, or delays in payment certification. Robust retrieval methods that

tolerate terminological variation while preserving numeric and parametric precision are therefore essential to practical construction workflows.

Attempts to automate this mapping process predominantly employed rule-based systems and traditional machine learning methods, which often rely on structured data sets and expert-driven input [7, 2]. Although these methods have shown limited success, they struggle with the domain-specific variability, nuanced language, and unstructured data prevalent in construction management [5, 8]. Advances in natural language processing (NLP), particularly transformer-based models such as BERT, have shown promise in addressing these challenges by capturing semantic variability and contextual relationships [9, 10]. However, issues such as data sparsity, domain-specific jargon, and computational resource demands persist [3, 11].

This study addresses these challenges through a systematic comparative evaluation of lexical, neural and hybrid retrieval methods for a large-scale construction catalog in the railway infrastructure domain. Using short-format catalog descriptions as queries and long-format entries as retrieval targets, we simulate realistic scenarios in which descriptions are shortened, paraphrased, or reformulated. The task is to retrieve the exact matching item among nearly forty thousand candidates from a single parametric sub-category, a setting where small lexical or numeric differences can determine success or failure.

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Figure 1 summarizes the retrieval setting considered in this study, from short catalog descriptions, used as proxies for practitioner-oriented queries, to the identification of the corresponding detailed construction item.

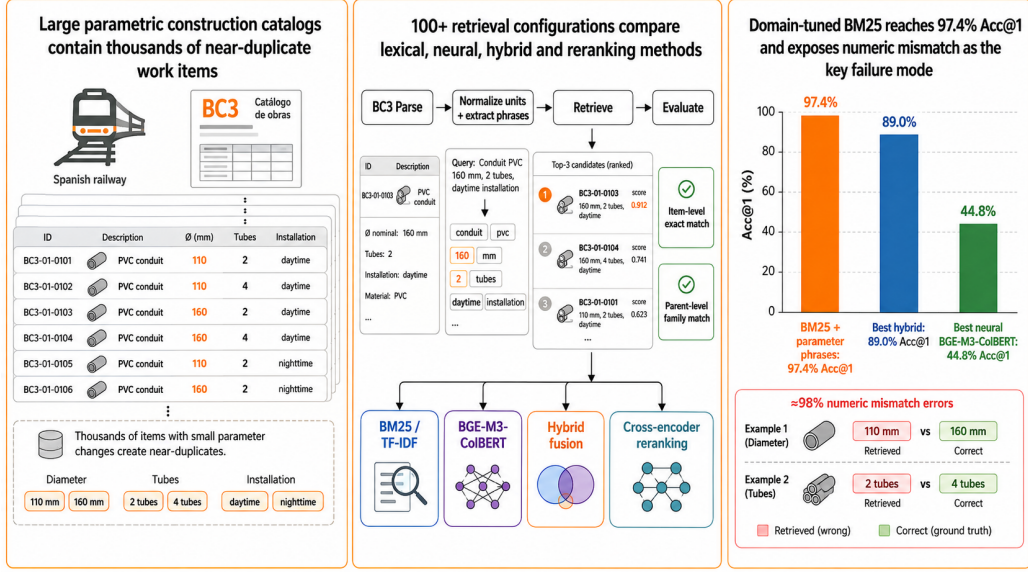


Figure 1: Overview of the construction catalog retrieval task evaluated in this study. Short catalog descriptions are used as proxy practitioner-oriented queries to retrieve the exact matching item from a large parametric catalog, where near-duplicate entries may differ only in technical or numeric attributes.

Our contributions are fourfold:

- We present a systematic comparison of lexical, sparse, dense, and hybrid retrieval methods, including pseudo-relevance feedback and cross-encoder reranking, in a precision-critical technical domain.
- We introduce an evaluation protocol that incorporates multiple metrics, dual target definitions (item- and parent-level), and significance testing, ensuring a rigorous and reproducible assessment.
- We conduct detailed error analysis, identifying numeric mismatches and near-duplicate confusions as dominant error modes, thus offering insights for future research on retrieval in structured domains.
- We create and release a structured benchmark dataset derived from the OEB subcategory of ADIF’s parametric railway catalog — a homogeneous yet challenging testbed of nearly 40,000 items exhibiting high parametric variability — including custom parsing tools to convert the

141 raw BC3 catalog format into a standardized retrieval evaluation dataset
142 [12].

143 The rest of this paper is organized as follows: Section 2 reviews prior work
144 on construction classification and retrieval methods. Section 3 describes the
145 ADIF dataset, preprocessing pipeline, and the 100 retrieval configurations
146 evaluated. Section 4 presents results showing that optimized lexical methods
147 consistently outperform neural and hybrid approaches at the item level, while
148 neural models remain competitive at the parent level. Section 5 discusses
149 the implications of these findings for construction catalog deployment, and
150 Section 6 concludes with directions for future research.

151 2. Prior Work

152 2.1. Construction Classification Systems and Standards

153 Construction work breakdown structures present a fundamental tax-
154 onomic challenge that distinguishes this domain from general informa-
155 tion retrieval applications. National and organizational classification sys-
156 tems—including MasterFormat, UniFormat, OmniClass, and infrastructure-
157 specific frameworks such as the ADIF (Spanish National Railway Manager)
158 railway catalog ADIF’s BPA Catalog used in this study—impose hierarchical
159 structures where items are organized by trade, material, location, or function
160 [13, 14]. These parametric catalogs contain thousands to tens of thousands
161 of work items characterized by structured attributes (dimensions, materials,
162 performance specifications, unit costs) alongside free-text descriptions that
163 exhibit substantial terminological variability. The ADIF catalog examined
164 here, for instance, contains approximately 47,500 distinct work items span-
165 ning civil works, track infrastructure, electrification, and signaling systems.

166 The vocabulary mismatch problem is particularly acute in construc-
167 tion contexts. Practitioners use regional terminology, abbreviations, brand
168 names, and informal descriptors that diverge significantly from standardized
169 catalog entries. A single concept—such as a reinforced concrete beam of spe-
170 cific dimensions—may appear under dozens of lexical variants across project
171 documents, specifications, and bills of quantities. This heterogeneity is fur-
172 ther compounded in multilingual settings and when integrating catalogs from
173 different organizations.

174 Prior work has addressed this complexity through ontology-based ap-
175 proaches. The Industry Foundation Classes (IFC) and International Frame-

work for Dictionaries (IFD) provide semantic frameworks for mapping between classification systems and enabling interoperability across software platforms. [15] developed natural-language retrieval engines for BIM object databases using vector-space models enhanced with domain ontologies, establishing a paradigm where lexical retrieval is augmented with explicit semantic expansion. [16] proposed Enhanced Explicit Semantic Analysis for construction product model retrieval, leveraging concept spaces derived from construction knowledge bases to bridge vocabulary gaps. [17] extended this work with multi-scale BIM information retrieval using hierarchical structure modeling based on IFC. These foundational approaches established that effective retrieval in construction requires explicit handling of domain semantics, though they predate the neural revolution in information retrieval and have not been systematically compared against modern dense retrieval methods.

2.2. Automated Classification and Retrieval in Construction

The application of natural language processing and machine learning to construction document processing has grown substantially over the past decade. Early work by [7] applied text classification to organize construction project documents, while subsequent studies extended these methods to bills of quantities, specifications, and regulatory texts. Systematic reviews by [18] and [3] document the evolution from rule-based and bag-of-words approaches through support vector machines to contemporary transformer-based methods.

Most published work in construction NLP addresses category-level classification rather than item-level retrieval. [9] applied deep learning to construction specification classification; [11] developed automated text classification pipelines for public procurement documents using support vector machines and later neural approaches. Jacques de Sousa et al. recently reviewed automation approaches for the budgeting phase, finding that text classification methods achieve reasonable accuracy for category assignment but struggle with fine-grained item matching.

Critically, classification accuracy degrades as taxonomic granularity increases. [7] reported 95.88% accuracy for top-level MasterFormat divisions but substantially lower performance for fine-grained subcategories. Similarly, [1] achieved 92.4% accuracy at the chapter level of work breakdown structures, with performance declining for detailed categorization. [10] explored bidirectional LSTMs and Temporal Convolutional Networks for

212 short construction text classification, demonstrating that even modern se-
213 quence models struggle with the brevity and density of catalog-style de-
214 scriptions. These findings underscore that the granularity required for item-
215 level retrieval—distinguishing among tens of thousands of near-identical en-
216 tries—exceeds the capabilities demonstrated in existing classification studies.

217 *2.3. Technical Catalog Retrieval: Methods and Cross-Domain Evidence*

218 Information retrieval for technical catalogs has received extensive atten-
219 tion in e-commerce and manufacturing domains, offering methodological in-
220 sights transferable to construction applications. The fundamental architec-
221 tural choice—lexical versus neural retrieval—has been systematically evalu-
222 ated in product search contexts with findings directly relevant to parametric
223 construction catalogs.

224 Lexical methods, exemplified by BM25 and its variants, remain remark-
225 ably competitive for technical domains. [19] compared BM25 against seman-
226 tic embedding methods on governmental product catalogs and electronic in-
227 voices, finding that lexical retrieval achieved faster identification of relevant
228 products while semantic methods improved ranking within retrieved sets.
229 This complementarity motivates hybrid architectures. [20] demonstrated
230 that sparse TF-IDF representations with extreme multi-label classification
231 (the PECOS framework) achieved superior recall and latency on catalogs
232 exceeding 100 million items compared to dense neural embeddings trained
233 on the same data. [21] showed that character-level TF-IDF outperformed
234 word-level variants for SKU search with abbreviation-heavy queries—a find-
235 ing directly applicable to construction catalogs where alphanumeric codes
236 and abbreviated specifications are prevalent. These results suggest that tech-
237 nical identifiers, part numbers, and precise parametric specifications benefit
238 from exact matching that dense representations may inadvertently blur.

239 Neural approaches have nonetheless advanced substantially in product
240 retrieval. Dual-encoder architectures trained on large-scale click logs demon-
241 strate strong performance when vocabulary mismatch is substantial and
242 paraphrase recognition is required. [22] at Amazon introduced semantic
243 product search using bi-encoder transformers, achieving significant improve-
244 ments in recall over lexical baselines. [23] developed structured matching
245 modules that encode multi-field product information—titles, descriptions,
246 dimensions, metadata—into unified representations that respect the hetero-
247 geneous structure of catalog entries. The CHARM architecture [24] proposed
248 cascading hierarchical attention to align query tokens with specific product

249 fields. More recent work explores learned sparse retrieval, where neural net-
250 works generate interpretable term expansions that combine lexical precision
251 with semantic generalization. [25] showed that hybrid encoders jointly learn-
252 ing dense and sparse representations can match cross-encoder effectiveness
253 while maintaining first-stage retrieval efficiency—a critical consideration for
254 production systems with latency constraints.

255 Relevant cross-domain work also addresses procurement and product
256 standardization. The TPDR system [26] developed for intermediary procure-
257 ment—including construction-related purchases—employs a two-stage archi-
258 tecture: dual transformer encoders for initial retrieval followed by syntactic
259 reranking. This hybrid approach achieved substantial gains over purely lex-
260 ical or purely semantic baselines when matching noisy, abbreviated initial
261 specifications to standardized product descriptions. Cross-lingual product
262 matching systems using hierarchical metric learning [27] further demon-
263 strate how taxonomy-aware training objectives can distinguish exact dupli-
264 cates from substitutes and unrelated items—a distinction critical for con-
265 struction catalogs where material grades or dimensional variants carry sig-
266 nificant cost implications.

267 *2.4. Research Gap and Contribution*

268 Despite the maturity of retrieval methods in adjacent domains, no system-
269 atic benchmark exists for evaluating lexical, neural, and hybrid approaches
270 on construction parametric catalogs. The AEC-specific literature remains
271 dominated by ontology-enhanced lexical methods developed before the trans-
272 former era, with limited adoption of the dense retrieval and learned sparse
273 techniques that have transformed e-commerce search. Meanwhile, the prod-
274 uct search literature, while methodologically sophisticated, lacks evaluation
275 on the particular characteristics of construction work items: dense parametric
276 specifications with numeric constraints, deep hierarchical classification struc-
277 tures, and domain-specific terminology in languages beyond English. This
278 gap leaves practitioners without empirical guidance when selecting retrieval
279 architectures for construction applications.

280 This study addresses that gap through a systematic comparison of re-
281 trieval methods on a large-scale Spanish railway construction catalog. We
282 evaluate classical lexical baselines (BM25 with domain-specific preprocessing
283 including Spanish stemming and stopword handling), neural dense retrievers
284 (multilingual sentence transformers fine-tuned for semantic similarity), and

285 hybrid configurations that combine lexical and neural signals. Our experi-
 286 mental design encompasses 100 retrieval configurations across realistic query
 287 scenarios reflecting practitioner search behavior. The findings provide em-
 288 pirical guidance for practitioners implementing catalog retrieval systems and
 289 contribute the first rigorous benchmark for information retrieval on paramet-
 290 ric construction work items, demonstrating conditions under which carefully
 291 tuned lexical methods remain competitive with—or superior to—neural al-
 292 ternatives.

293 **3. Methodology**

294 *3.1. Dataset Characteristics and Processing*

295 *3.1.1. Dataset Scope and Selection*

296 The experimental dataset derives from ADIF’s parametric price cata-
 297 log for Spanish railway infrastructure. While the complete catalog spans
 298 ten technical chapters (Architecture, Civil Works, Control Systems, Energy,
 299 Environmental Management, Protection and Security, Safety and Health,
 300 Telecommunications, and Track), this study focuses on the OEB subcategory
 301 within the Civil Works chapter, which addresses platform system infrastruc-
 302 ture such as trenches, conduits, and piping systems, providing a controlled
 303 yet demanding testbed for retrieval evaluation. Extending the benchmark
 304 to additional chapters remains a priority for future work, as discussed in
 305 Section 5.

306 The OEB subcategory provides an ideal testbed for retrieval evaluation as
 307 it exhibits high parametric variability while maintaining manageable scope
 308 for comprehensive analysis. Through parametric expansion of base templates
 309 across combinations of pipe materials, diameters (110mm, 160mm, 200mm),
 310 quantities (2, 4, 6, 8, 12 tubes), terrain types (normal, rocky, under tracks,
 311 road crossing, platform, attached, ballast), and work shift specifications (day-
 312 time, nighttime, exceptional conditions), the subcategory generates 47513
 313 distinct item variants [12].

314 Each of these item variants belongs to a hierarchical structure: - The
 315 *item-level* contains the fully specified variant, including all parameters (e.g.,
 316 “6 PVC tubes, 160mm, under tracks, nighttime”). - The *parent-level* refers to
 317 the shared base template that groups multiple such variants under a common
 318 parametric definition.

319 Table 1 presents key characteristics of the processed dataset.

Table 1: OEB Subcategory Dataset Statistics

Characteristic	Value
Total unique items	47,513
Base templates	30
Average expansion factor	1583.8
<i>Short-Format description</i> (<i>‘resumen’</i>) (query)	
Mean length (words)	19.7
Median length (words)	20
Std deviation (words)	2.4
<i>Long-Format description</i> (<i>‘texto’</i>) (target)	
Mean length (words)	84.5
Median length (words)	83
Std deviation (words)	15.0
Queries with numeric parameters	47,430 (99%)
Queries without numeric parameters	83 (0%)
Single-item templates (=1 variants)	25 (0%)

Note: ‘resumen’ and ‘texto’ are the field names in the original Spanish dataset.

To illustrate the dataset structure, Figure 2 presents a translated example from the original Spanish corpus. A template (e.g., OEB020\$) defines a base construction concept and several configurable parameters. Each unique combination of parameter values produces a distinct item, with both a concise short-format description used as the query and a detailed long-format description serving as the retrieval target.

3.1.2. BC3 Format Processing

ADIF distributes its catalog in BC3 format (Base de Costes de la Construcción), the standard exchange format established by FIEBDC, a Spanish industry consortium that maintains data interchange standards for construction cost databases. We developed a custom parsing pipeline to transform the BC3 specification into structured JSON suitable for retrieval experiments.

The parsing pipeline handles three key transformations:

1. **Parametric Expansion:** The BC3 format encodes base items with parameter placeholders (e.g., \No_PIPES\, \TERRAIN_TYPE\). The parser extracts parameter definitions, generates all valid combinations, and instantiates complete item descriptions by substituting parameter values into templates.

Template: OEB020\$ — Concrete-encased PVC conduit, 110 mm	
Parameters:	
No. of pipes	2
Terrain type	Normal
Work shift	Daytime
Maintenance window	≥ 5 hours
Execution conditions	High volume
Short-format (<i>resumen</i>):	
Concrete-encased conduit, 2 pipes, PVC 110 mm, normal terrain. (Daytime / ≥ 5 h / High volume)	
Long-format (<i>texto</i>):	
Concrete-encased conduit installation of 2 PVC pipes, 110 mm diameter, in any terrain except rock, including trench backfilling and compaction, supply and installation of pipes and HE-20 concrete (non-vibrated), conduit testing, transport and removal of materials to worksite. Work shift: Daytime. Maintenance window: ≥ 5 hours. Execution conditions: High volume.	

Figure 2: Example item from the OEB dataset (translated from Spanish). Each template expands into multiple items by varying parameter combinations.

2. **Dual-Text Extraction:** Each item contains two text fields: *short-format* (condensed summary with parametric codes) and *long-format* (detailed technical specification). The parser extracts both fields while preserving their association through unique item identifiers.
3. **Metadata Preservation:** The parser retains hierarchical information including chapter codes, base template identifiers (first 6 characters), and parametric family relationships to enable parent-level evaluation.

The complete parsing pipeline implementation is available in our code repository [28].

3.1.3. Text Preprocessing

All text fields undergo standardized preprocessing before indexing and retrieval. The preprocessing pipeline applies the following transformations:

Normalization.

- Lowercase conversion for case-insensitive matching
- Whitespace standardization (multiple spaces $\rightarrow$ single space)

- 355 • Punctuation preservation except for sentence-final periods
- 356 • Numeric normalization (e.g., “1,256.84” for “1256,84”)
- 357 • Accented character retention for proper Spanish processing

358 *Tokenization.* Different retrieval methods receive appropriately tokenized input:

- 359 • **Word tokens:** Regex pattern `\w+` for lexical methods, yielding
360 sequences like `["conduit", "concrete-cased", "2", "pipes", "pvc",`
361 `"110", "mm"]`
- 362 • **Character n-grams:** Overlapping sequences of length 3, 4, and 5 for
363 character-based methods (e.g., `"pvc" → ["pvc", "#pv", "pvc", "vc#"]`)
- 364 • **Subword tokens:** Pretrained tokenizers for neural methods (WordPiece
365 for BERT-based models, SentencePiece for others)

366 *Parameter-Based Phrase Identification.* We leverage the catalog’s parametric
367 structure to identify domain-specific multi-word expressions. From the BC3 pa-
368 rameter definitions, we extract 127 multi-word parameter values (e.g., terrain
369 types, work conditions, installation contexts) that function as atomic units in
370 the classification system.

371 The preprocessing pipeline evaluated two phrase handling strategies:

- 372 • **Additive:** Phrases supplement unigrams (“road”, “crossing”,
373 “road_crossing”)
- 374 • **Replacement:** Phrases substitute component words (“road_crossing”
375 only)

376 This domain-specific phrase handling enables retrieval methods to align with
377 the catalog’s inherent parametric vocabulary.

378 *Numeric Feature Extraction.* The pipeline identifies and normalizes integer quan-
379 tities, decimal measurements with units, numeric ranges, and percentages.

380 Extracted numeric features enable exact numeric matching as a complementary
381 retrieval signal (Section 3.2.4).

382 3.2. Retrieval System Configurations

383 We evaluate 100 distinct retrieval configurations organized into four method
384 families. Table 2 summarizes the configuration space.

Table 2: Overview of Evaluated Retrieval Configurations

Method Family	Configs	Key Variations
Lexical (BM25 & TF-IDF)	67	Parameter tuning, tokenization, phrase handling
Pseudo-Relevance Feedback	16	RM3 and Rocchio with various feedback parameters
Neural Dense Retrieval	7	E5, GTE, BGE-M3 models with different settings
Hybrid & Reranking	10	Fusion methods and cross-encoder reranking
Total	100	

3.2.1. Lexical Methods: BM25 and TF-IDF

BM25 Implementation. We implement BM25 using Pyserini v0.21.0 [29], which provides efficient indexing and retrieval through the Lucene backend. We evaluate three tokenization variants, each with its own parameter grid:

- **BM25-unigram:** Standard word tokenization
 $k_1 \in \{0.6, 0.8, 1.0, 1.2, 1.4\}$, $b \in \{0.20, 0.35, 0.50, 0.65, 0.80\}$ (25 combinations)
- **BM25-unibigram:** Word unigrams + bigrams
 $k_1 \in \{0.8, 1.0, 1.2, 1.4, 1.6\}$, $b \in \{0.35, 0.50, 0.65, 0.80, 0.95\}$ (25 combinations)
- **BM25-unigram-params:** Word tokens + parameter-derived phrases
 $k_1 \in \{0.6, 0.8, 1.0, 1.2\}$, $b \in \{0.35, 0.50, 0.65\}$ (12 combinations)

This yields 62 total configurations. Documents and queries undergo identical preprocessing before indexing. Index construction uses default Lucene analyzers with custom stopwords lists removed (domain-specific terms like “of”, “in” carry meaning in construction context).

TF-IDF Implementation. TF-IDF baselines use scikit-learn v1.2.2 [30] with the following configurations:

- **TF-IDF-unigram:** Word unigrams with standard stopwords removal
- **TF-IDF-unigram-nonstop:** Word unigrams without stopwords removal
- **TF-IDF-char:** Character n-grams ($n \in \{3, 4, 5\}$) with word boundary markers

- 407 • **TF-IDF-phrases-add**: Word unigrams augmented with parameter-derived
408 phrase tokens (e.g., “High volume”)
- 409 • **TF-IDF-phrases-replace**: Multi-word parameter phrases replaced by sin-
410 gle tokens

411 All TF-IDF variants use cosine similarity for ranking with L2-normalized vectors.

412 3.2.2. Pseudo-Relevance Feedback Methods

413 We evaluate two PRF methods using the same parameter grid: feedback
414 documents $M \in \{5, 10\}$, expansion terms $N \in \{20, 40\}$, and feedback weight
415 $\beta \in \{0.5, 0.7\}$, yielding 8 configurations each (the two weakest $\beta=0.7$, $N=40$ vari-
416 ants are omitted for brevity; full results in the repository).

417 *RM3*. It uses Pyserini’s built-in PRF module with BM25 as the base retriever,
418 where β controls interpolation between the original query and the relevance model.

419 *Rocchio*. It uses scikit-learn with TF-IDF representations, where β weights the
420 feedback centroid. No negative feedback is used ($\gamma = 0$) due to the single-relevant-
421 item setting.

422 3.2.3. Neural Dense Retrieval Methods

423 All neural models are used in zero-shot mode without fine-tuning on construc-
424 tion text, testing their transfer capabilities to domain-specific Spanish technical
425 language.

426 *E5 Model*. We evaluate `intfloat/multilingual-e5-base` (278M parameters,
427 768 dimensions) using the sentence-transformers library v2.2.2 [31]. Following the
428 model documentation, queries receive the prefix “query: ” and documents receive
429 “passage: ” before encoding. Embeddings are L2-normalized before similarity
430 computation.

431 *Spanish Sentence Similarity*. We evaluate `hiiasid/sentence_similarity_spanish_es`,
432 a model specifically fine-tuned for Spanish semantic similarity tasks, to compare
433 against multilingual alternatives.

434 *GTE Model*. We evaluate `Alibaba-NLP/gte-multilingual-base` with two query
435 formats:

- 436 • **GTE-direct**: Standard “query:” prefix
- 437 • **GTE-instruct**: Spanish instruction prefix: “query: Lookup for the most
438 similar document to this query: [query]”

439 *BGE-M3 Model.* BGE-M3 (BAAI/bge-m3, 568M parameters) supports three re-
 440 trieval modes within a unified architecture [32]:

- 441 • **BGE-M3-dense:** Single 1024-dimensional vector per text
- 442 • **BGE-M3-sparse:** Learned sparse weights (SPLADE-style)
- 443 • **BGE-M3-colbert:** Token-level embeddings with ColBERT-style late in-
 444 teraction

445 The ColBERT mode computes relevance as the sum of maximum token similarities.
 446 All BGE-M3 modes use the same encoder with mode-specific output layers.

447 3.2.4. Hybrid Fusion and Reranking

448 The fusion process consisted of three sequential steps applied to each query
 449 independently.

450 First, retrieval scores from each ranker were normalized using min-max scaling
 451 to ensure comparability across heterogeneous scoring functions. For a given query
 452 q , document d and ranker r , each document score $s_{d,r}$ was transformed as:

$$s'_{d,r} = \frac{s_{d,r} - \min_r(s)}{\max_r(s) - \min_r(s)} \quad (1)$$

453 Second, we applied numeric parameter bonuses to account for matches between
 454 query parameters and document metadata. Each document received an additional
 455 weighted reward based on exact parameter matches (β_{exact}) and partial parameter
 456 overlaps (β_{overlap}). These bonuses were added to the normalized scores before
 457 fusion.

458 Third, the adjusted scores from all retrievers were combined using weighted
 459 linear combination (COMBSUM):

$$s_{\text{final}}(d) = \sum_{r \in R} w_r \cdot s'_{d,r} \quad (2)$$

460 where w_r represents the fusion weight for ranker r , and $\sum_r w_r = 1$.

461 To identify optimal fusion configurations, we conducted a systematic paramete-
 462 ter sweep over the weight space for several hybrid strategies: (1) three-way fusions
 463 combining BM25 with one BGE-M3 variant and character-level TF-IDF, and (2)
 464 a five-way fusion incorporating BM25, all three BGE-M3 variants (dense, sparse,
 465 ColBERT), and TF-IDF. For each configuration, we evaluated combinations of
 466 $\beta_{\text{exact}} \in \{0.0, 0.05, 0.1\}$ and $\beta_{\text{overlap}} \in \{0.0, 0.05\}$ across a grid of weight distribu-
 467 tions. The optimal configuration for each strategy was selected based on MRR@10
 468 performance on the test set.

469 *Cross-Encoder Reranking.* We employ a two-stage cascade: efficient first-stage
 470 retrieval followed by expensive cross-encoder reranking of top candidates. Cross-
 471 encoder implementation uses:

- 472 • Model: `cross-encoder/ms-marco-MiniLM-L-6-v2` (22M parameters)
- 473 • Input format: `[CLS] query [SEP] document [SEP]`
- 474 • Reranking depth: Top 20, 50, or 100 candidates from first stage
- 475 • Base retrievers: BM25, E5-large, and hybrid configurations

476 The cross-encoder produces a relevance score for each query-document pair
 477 through joint encoding and classification head. We evaluate the performance gain
 478 versus computational cost across different reranking depths.

479 3.3. Evaluation Protocol

480 3.3.1. Retrieval Task Definition

481 The evaluation task simulates realistic construction catalog search: given a
 482 short-form description (*resumen*), retrieve the corresponding detailed specification
 483 (*texto*) from 47,513 candidates. Each query has exactly one correct target at the
 484 item level, though multiple items may share the same parametric parent.

485 We construct the evaluation set by:

- 486 1. A random sample of 16,590 from the *resumen* descriptions was selected as
 487 query inputs
- 488 2. All 47,513 *texto* descriptions were used as the retrieval corpus
- 489 3. Ground truth mappings were defined using shared item identifiers
- 490 4. The full catalog was retained as distractors-no filtering or candidate pruning
 491 was applied

492 The sample size (16,590) ensures 99% confidence with $\pm 1\%$ margin of error:

$$n = \frac{Z^2 \cdot p(1-p)}{E^2} = \frac{(2.576)^2 \cdot 0.5 \cdot 0.5}{(0.01)^2} = 16,590$$

493 where n is the required sample size, Z is the Z-score for the confidence level (2.576
 494 for 99%), $p = 0.5$ assumes maximum variance, and E is the margin of error (0.01).
 495 This ensures statistically robust comparisons across methods.

496 3.3.2. Dual-Target Evaluation

497 We implement two complementary evaluation perspectives:

498 *Item-Level (Exact Match)*. Requires retrieving the specific parametric variant. For
 499 query with ID “OEB020aabdcfa”, only the document with identical ID counts as
 500 correct. This strict criterion applies to scenarios requiring precise specifications
 501 such as material procurement or detailed cost estimation.

502 *Parent-Level (Family Match)*. Credits retrieval of any variant from the same
 503 parametric family, identified through shared 6-character prefixes. For query
 504 “OEB020aabdcfa”, any document with ID starting “OEB020” counts as correct.
 505 This relaxed criterion acknowledges scenarios where parametric siblings are func-
 506 tionally equivalent, such as preliminary estimation or category identification.

507 The dual-target approach enables nuanced analysis: methods may excel at
 508 family identification while struggling with variant-level disambiguation.

509 3.3.3. Evaluation Metrics

510 We employ five complementary metrics to assess retrieval quality:

- 511 1. **Accuracy@1 (Success@1)**: Proportion of queries where the correct item
 512 appears in rank 1. Critical for automated systems where only the top result
 513 is used.
- 514 2. **Recall@k**: Proportion of queries with correct item in top- k results. We
 515 report Recall@5 and Recall@10 as reasonable examination depths for human
 516 review.
- 517 3. **Mean Reciprocal Rank (MRR)**: Average of reciprocal ranks of first cor-
 518 rect result.
- 519 4. **Normalized Discounted Cumulative Gain (nDCG@10)**: Accounts
 520 for rank position using logarithmic discount
- 521 5. **Mean Average Precision (MAP)**: Averaged precision at each relevant
 522 result.

523 All metrics are computed using the ranx library v0.3.7 [33], which provides
 524 efficient implementations with built-in significance testing.

525 3.3.4. Query Subset Analysis

526 Performance analysis across query subsets reveals how methods handle different
 527 retrieval challenges. We partition the evaluation set into two categories:

- 528 • **HAS_NUM** (24,711 queries, 62%): Queries containing at least one nu-
 529 meric parameter (e.g., “2 tubos”, “110 mm”). Tests whether methods can
 530 match quantitative specifications.

- 531 • **NO_NUM** (15,136 queries, 38%): Queries without numeric content, con-
532 taining only textual descriptors. Isolates purely textual matching perfor-
533 mance.

534 Subset analysis identifies method strengths (e.g., neural models may excel on
535 NO_NUM while lexical methods dominate HAS_NUM).

536 3.3.5. Statistical Significance Testing

537 We employ paired bootstrap resampling to determine whether observed per-
538 formance differences reflect genuine method advantages rather than random vari-
539 ation [34].

540 For each system pair (A, B) , we compute per-query metric differences, generate
541 $B = 10,000$ bootstrap samples with replacement, and estimate p -values from the
542 empirical distribution of mean differences. We apply Bonferroni correction for
543 multiple comparisons and report improvements as significant only when $p < 0.05$
544 after correction.

545 3.3.6. Error Analysis

546 To understand failure modes, we analyze 100 randomly sampled errors from
547 each method family. Errors are categorized into:

- 548 1. **Numeric Mismatch:** Retrieved item differs from target in quantitative
549 parameters (e.g., “110 mm” vs “160 mm”, “2 tubos” vs “4 tubos”)
- 550 2. **Lexical Confusion:** High lexical overlap masks critical distinctions, par-
551 ticularly in work condition modifiers (e.g., standard vs exceptional shifts) or
552 scope descriptors (e.g., supply-only vs supply-and-installation)
- 553 3. **Near-Duplicate Error:** Correct item exists but parametric sibling ranks
554 higher due to high lexical similarity
- 555 4. **Ranking Failure:** Correct item retrieved but ranks beyond typical exami-
556 nation depth (position 11+)

557 Error distribution reveals systematic failure patterns that guide method selec-
558 tion for different deployment contexts.

559 3.4. Implementation Details

560 All experiments use Python 3.10 with Pyserini for BM25/RM3, scikit-learn for
561 TF-IDF/Rocchio, sentence-transformers and FlagEmbedding for neural models,
562 and ranx for evaluation [33]. Experiments run on servers with Intel i9-13900KF
563 CPUs and 4× NVIDIA RTX 4090 GPUs. For reproducibility, random seeds are
564 fixed, model versions pinned, and complete configurations stored in YAML files;
565 code and data are openly available at [28, 12].

4. Results

The results across all evaluated configurations converge on a consistent finding: lexical methods, when properly tuned with domain-specific parameter phrases, outperform neural and hybrid approaches at the item level by a substantial margin. The following subsections detail the performance of each method family, culminating in a cross-method comparison in Section 4.7.

4.1. Lexical Methods Comparative Analysis

The performance of all lexical retrieval configurations was evaluated at two hierarchical levels: parent-level and item-level retrieval. Figure 3 presents parent-level results, where the task is to retrieve the correct parent document, while Figure 4 shows item-level results for retrieving specific document items. Both evaluations used the same test set of 16,590 queries.

At the parent level (Figure 3), TF-IDF methods dominated, with TF-IDF-unigram_phrases_replace achieving the highest Acc@1 (0.990). BM25 variants showed more modest parent-level performance, ranging from 0.873 to 0.985.

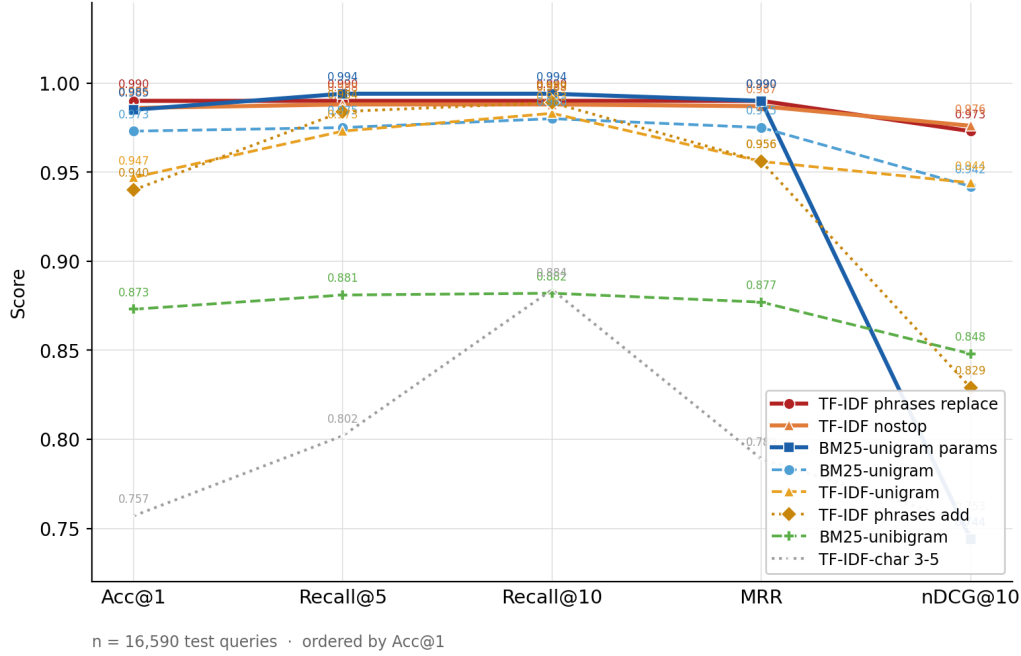


Figure 3: Performance comparison of lexical retrieval methods at parent level on the test set (n=16,590 queries). Methods are ranked by Accuracy@1.

At the item level (Figure 4), the ranking inverted. BM25-unigram_params achieved the highest Acc@1 (0.974), while TF-IDF methods dropped substantially (0.558–0.708). This 26.6 percentage point gap between the best BM25 and best TF-IDF demonstrates that document-length normalization is critical for fine-grained catalog discrimination.

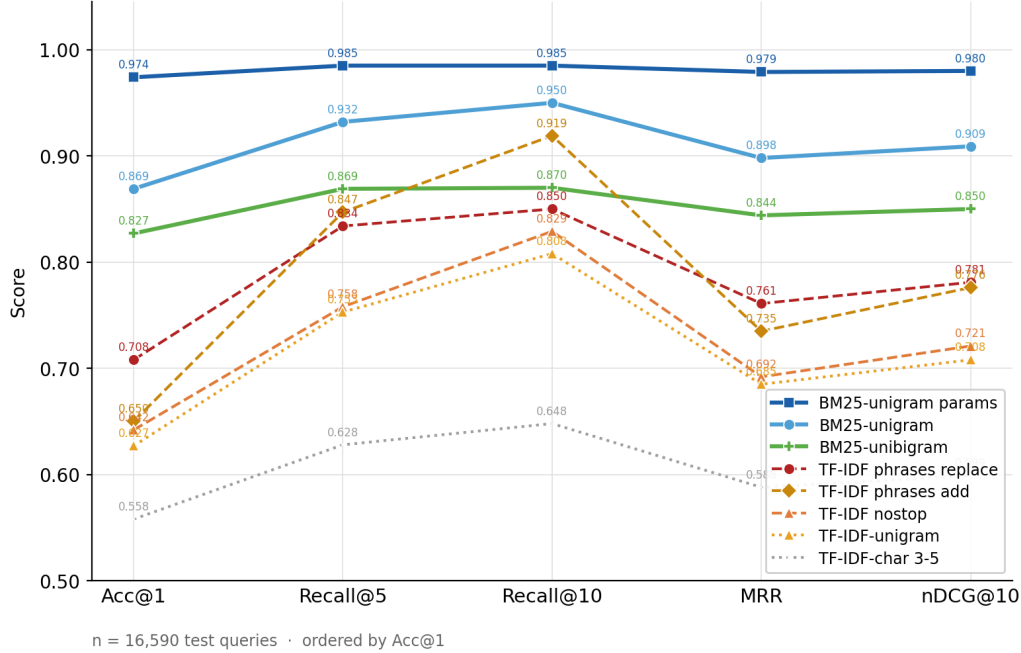


Figure 4: Performance comparison of lexical retrieval methods at item level on the test set (n=16,590 queries). Methods are ranked by Accuracy@1.

Figure 5 visualizes the sensitivity of item-level Acc@1 to the BM25 hyperparameters (k_1, b) across the three tokenization variants. The parameter-phrase variant dominates the entire explored grid, not only at its optimum, indicating that domain-aware phrase tokenization provides a structural advantage that is robust to hyperparameter choice. All three variants converge on a preference for low b (≈ 0.35), suggesting that aggressive length normalization is counterproductive in this catalog where short and long fields differ systematically rather than randomly.

Bootstrap significance testing (10,000 iterations, Bonferroni correction) confirmed all pairwise differences were statistically significant ($p < 0.003$).

Table 3 presents pairwise comparisons. BM25-unigram outperformed all TF-IDF variants by 16.1–31.1 percentage points, and exceeded BM25-unigram by 4.2 percentage points.

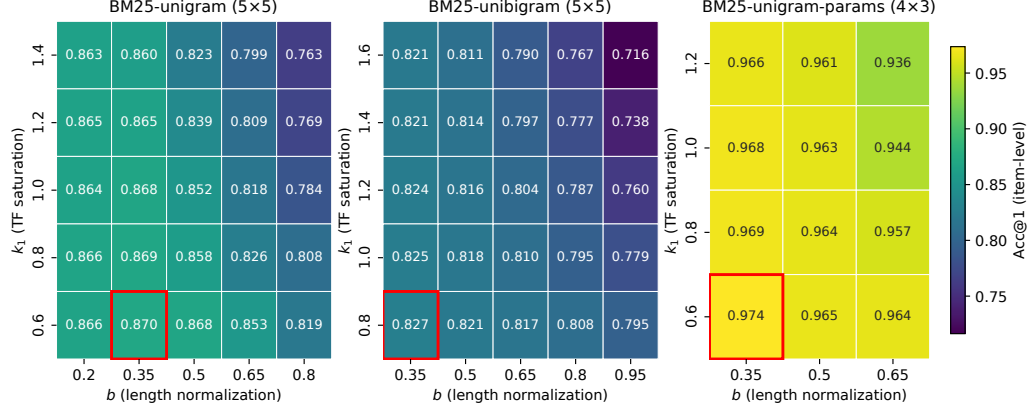


Figure 5: Acc@1 sensitivity to BM25 hyperparameters (k_1 , b) across three tokenization variants on the item-level retrieval task ($n = 16,590$ queries). A shared color scale highlights the systematic dominance of the parameter-phrase variant (right panel) across the explored grid. The optimum of each panel is highlighted with a red border.

Table 3: Pairwise statistical significance testing results for item-level lexical method performance. Bootstrap test with 10,000 iterations; p-values corrected using Holm-Bonferroni method. All comparisons show significant differences ($p < 0.003$).

Method 1	Method 2	Δ Acc@1	95% CI	p-value	Sig.
BM25-unigram	BM25-unigram	0.042	[0.036, 0.048]	< 0.001	***
BM25-unigram	TF-IDF-phrases_replace	0.161	[0.153, 0.169]	< 0.001	***
BM25-unigram	TF-IDF-phrases_add	0.219	[0.211, 0.227]	< 0.001	***
BM25-unigram	TF-IDF-nostop	0.227	[0.220, 0.235]	< 0.001	***
BM25-unigram	TF-IDF-unigram	0.243	[0.235, 0.250]	< 0.001	***
BM25-unigram	TF-IDF-char_3_5	0.311	[0.303, 0.319]	< 0.001	***
BM25-unigram	TF-IDF-phrases_replace	0.118	[0.112, 0.125]	< 0.001	***
BM25-unigram	TF-IDF-phrases_add	0.177	[0.170, 0.184]	< 0.001	***
BM25-unigram	TF-IDF-nostop	0.185	[0.178, 0.192]	< 0.001	***
BM25-unigram	TF-IDF-unigram	0.200	[0.193, 0.207]	< 0.001	***
BM25-unigram	TF-IDF-char_3_5	0.268	[0.261, 0.276]	< 0.001	***
TF-IDF-phrases_replace	TF-IDF-phrases_add	0.058	[0.053, 0.064]	< 0.001	***
TF-IDF-phrases_replace	TF-IDF-nostop	0.067	[0.061, 0.072]	< 0.001	***
TF-IDF-phrases_replace	TF-IDF-unigram	0.082	[0.076, 0.088]	< 0.001	***
TF-IDF-phrases_replace	TF-IDF-char_3_5	0.150	[0.143, 0.157]	< 0.001	***
TF-IDF-nostop	TF-IDF-phrases_add	0.008	[0.003, 0.014]	< 0.003	**
TF-IDF-unigram	TF-IDF-nostop	0.015	[0.010, 0.020]	< 0.001	***
TF-IDF-unigram	TF-IDF-phrases_add	0.023	[0.019, 0.027]	< 0.001	***

*** $p < 0.001$; ** $p < 0.01$ (Holm-Bonferroni corrected)

598 Stratification by query type (Table 4) revealed that BM25-params achieved
 599 perfect accuracy on non-numeric queries (0.17% of test set) while maintaining
 600 0.974 on numeric ones. TF-IDF variants showed the inverse: perfect non-numeric
 601 but weaker numeric performance (0.558–0.708).

Table 4: Item-level performance stratification by query type. The test set comprises 16,076 queries with numbers (99.83%) and 29 queries without numbers (0.17%).

Method	Acc@1		Recall@10	
	Numeric	Non-numeric	Numeric	Non-numeric
BM25-unigram_params (k1=0.60, b=0.35)	0.974	1.000	0.985	1.000
BM25-unigram (k1=0.80, b=0.35)	0.869	0.931	0.950	1.000
BM25-unibigram (k1=0.80, b=0.35)	0.826	0.966	0.870	0.966
TF-IDF-unigram_phrases_replace	0.708	1.000	0.850	1.000
TF-IDF-unigram_phrases_add	0.649	1.000	0.919	1.000
TF-IDF-unigram_nostop	0.641	1.000	0.829	1.000
TF-IDF-unigram	0.626	1.000	0.807	1.000
TF-IDF-char_3_5	0.558	1.000	0.647	1.000

602 Based on item-level results, BM25 with optimized parameters (k1=0.60,
 603 b=0.35) was selected as the best-performing lexical method for subsequent pseudo-
 604 relevance feedback experiments, demonstrating superior item-level performance
 605 across all evaluation metrics with statistical significance.

606 4.2. Pseudo-Relevance Feedback Methods

607 Pseudo-relevance feedback (PRF) methods were evaluated to assess their effec-
 608 tiveness in enhancing retrieval performance beyond the base lexical retriever. Two
 609 PRF approaches were tested: RM3 (Relevance Model 3) and Rocchio feedback.
 610 Both methods leveraged BM25-unigram (k1=0.80, b=0.35) as their base retriever,
 611 which achieved Acc@1 of 0.869, Recall@5 of 0.932, Recall@10 of 0.950, MRR of
 612 0.898, and nDCG@10 of 0.909 at the item level.

613 4.2.1. RM3 and Rocchio Performance

614 Six configurations each were evaluated for RM3 and Rocchio, varying feed-
 615 back documents ($M \in \{5, 10\}$), expansion terms ($N \in \{20, 40\}$), and interpolation
 616 weights. Tables 5 and 6 present detailed results. RM3 showed remarkable sta-
 617 bility across configurations (Acc@1: 0.8693–0.8699), while Rocchio performance
 618 degraded notably with N=40 terms (Acc@1 dropping from 0.857 to 0.837).

619 4.2.2. Comparative Analysis and Statistical Significance

620 Table 7 presents the performance comparison between the best PRF methods
 621 and their base retriever. Statistical significance testing was conducted using paired

Table 5: Performance of RM3 configurations at item level on the test set (n=16,590 queries). All configurations use BM25-unigram as the base retriever.

M	N	λ	MRR	nDCG@10	Acc@1
5	20	0.5	0.8983	0.9093	0.8699
5	20	0.7	0.8982	0.9092	0.8698
5	40	0.5	0.8982	0.9092	0.8699
10	20	0.5	0.8983	0.9093	0.8699
10	20	0.7	0.8978	0.9089	0.8693
10	40	0.5	0.8982	0.9092	0.8699

Table 6: Performance of Rocchio configurations at item level on the test set (n=16,590 queries). All configurations use BM25-unigram as the base retriever.

M	N	β	MRR	nDCG@10	Acc@1
5	20	0.5	0.8875	0.9000	0.8565
5	20	0.7	0.8874	0.8999	0.8563
5	40	0.5	0.8718	0.8877	0.8373
10	20	0.5	0.8873	0.9000	0.8559
10	20	0.7	0.8872	0.8999	0.8559
10	40	0.5	0.8713	0.8873	0.8361

bootstrap resampling with 10,000 iterations and $\alpha=0.05$, with multiple testing correction using the Holm–Bonferroni method.

Table 7: Comparison of best PRF methods versus base retriever at item level. Statistical significance was assessed using bootstrap resampling (10,000 iterations) with Holm–Bonferroni correction.

Method	MRR	nDCG@10	Acc@1	Significance
Base BM25-unigram	0.898	0.909	0.869	—
Best RM3 (M=10, N=20, $\lambda=0.5$)	0.898	0.909	0.870	n.s.
Best Rocchio (M=5, N=20, $\beta=0.5$)	0.888	0.900	0.857	$p < 0.05$

Table 7 summarizes the comparison. RM3 showed no statistically significant improvement over the baseline (95% CI: [-0.0005, 0.0021]). Rocchio produced significant degradation ($p < 0.05$, $\Delta = 1.26\%$), suggesting that query expansion is ill-suited for this domain where precise numeric matching outweighs semantic expansion.

The strong baseline BM25 performance leaves minimal room for improvement through query expansion, indicating limited value of PRF for this technical retrieval task.

4.3. Neural Dense Retrieval Methods

This subsection presents the comparative performance of neural dense retrieval models across three model families: E5, GTE, and BGE-M3. All models were evaluated in zero-shot mode on the same test set of 16,590 queries at both item-level and parent-level retrieval tasks.

4.3.1. E5 Model Family Performance

Table 8 presents the performance comparison across E5 model variants. The multilingual-e5-base model achieved item-level Acc@1 of 0.134 (13.4%), meaning only 1 in 7.5 queries successfully retrieved the correct item as the top result. At the parent level, E5-large demonstrated substantially stronger performance with Acc@1 of 0.982 (98.2%).

The E5 model demonstrated dramatically different performance between hierarchical levels, with Acc@1 improving 7.3-fold from 13.4% at item-level to 98.2% at parent-level. This pattern suggests that while E5 embeddings effectively capture high-level semantic similarity for parent document matching, they struggle to discriminate between fine-grained item variations within the same parent category.

Stratification by query type revealed a critical limitation: queries without numeric parameters achieved Acc@1 of 0.875 versus 0.133 for numeric queries — a pattern that persists across all neural model families.

Table 8: Performance comparison of E5 model variants. All models evaluated on 16,589 queries (test set).

Model	Acc@1	Recall@5	Recall@10	MRR	nDCG@10
<i>Item-Level Retrieval</i>					
E5-large	0.134	0.376	0.518	0.251	0.303
<i>Parent-Level Retrieval</i>					
E5-large	0.982	0.993	0.996	0.987	0.969

651 4.3.2. GTE Model Family Performance

652 Table 9 presents results for both GTE model variants: the base Alibaba-
653 NLP/gte-multilingual-base model and the instruction-following Alibaba-NLP/gte-
654 multilingual-base instruct variant. Contrary to expectations that the larger
655 instruction-tuned model would demonstrate superior performance, the base GTE-
656 large model outperformed the instruct variant at the parent level while showing
657 marginal improvement at the item level.

Table 9: Performance comparison of GTE model variants with different query formulations.

Model	Acc@1	Recall@5	Recall@10	MRR	nDCG@10
<i>Item-Level Retrieval</i>					
GTE-large-en-v1.5	0.013	0.044	0.066	0.030	0.036
GTE-Qwen2-instruct	0.014	0.049	0.074	0.034	0.040
<i>Parent-Level Retrieval</i>					
GTE-large-en-v1.5	0.837	0.941	0.961	0.884	0.701
GTE-Qwen2-instruct	0.697	0.938	0.978	0.798	0.651

658 Both GTE variants achieved the poorest item-level performance among neural
659 families (Acc@1: 1.3–1.4%). At the parent level, GTE-large reached 0.837, but the
660 instruction-tuned variant unexpectedly degraded to 0.697, suggesting that explicit
661 prompts introduce noise in structured technical retrieval tasks.

662 Non-numeric queries showed substantial improvements (GTE-Qwen2: 37.9%
663 vs 1.4% overall), reinforcing the numeric precision challenge observed with E5.

664 4.3.3. BGE-M3 Multi-Representation Architecture

665 Table 10 presents performance across three retrieval modes of the BGE-M3
666 unified architecture: dense single-vector, learned sparse representations, and multi-
667 vector ColBERT-style interactions. The multi-vector mode demonstrated substan-
668 tially superior item-level performance compared to both single-vector approaches.

Table 10: Performance comparison of BGE-M3 retrieval modes. All modes use the same base encoder (568M parameters) with mode-specific output layers.

Mode	Acc@1	Recall@5	Recall@10	MRR	nDCG@10
<i>Item-Level Retrieval</i>					
BGE-M3-dense	0.127	0.349	0.464	0.233	0.278
BGE-M3-sparse	0.125	0.401	0.535	0.252	0.309
BGE-M3-colbert	0.448	0.753	0.785	0.580	0.630
<i>Parent-Level Retrieval</i>					
BGE-M3-dense	0.960	0.984	0.993	0.971	0.906
BGE-M3-sparse	0.994	0.999	1.000	0.996	0.983
BGE-M3-colbert	0.997	0.999	1.000	0.998	0.958

At the item level, BGE-M3-ColBERT achieved Acc@1 of 0.448 (44.8%), dramatically outperforming the dense mode at 0.127 (12.7%) and sparse mode at 0.125 (12.5%). This represents a 3.5-fold improvement over both single-vector approaches. The ColBERT multi-vector approach also substantially outperformed all E5 and GTE variants, achieving 3.3-fold higher Acc@1 than E5-large (0.134) and 34-fold higher Acc@1 than GTE-large (0.013).

Dense and sparse modes converged at nearly identical accuracy (12.5–12.7%), suggesting that single-vector compression encounters similar limitations regardless of representation strategy.

At the parent level, all three modes demonstrated strong performance exceeding 96% Acc@1, with the ColBERT mode achieving the highest accuracy at 0.997 (99.7%), followed closely by sparse at 0.994 (99.4%) and dense at 0.960 (96.0%). The narrow ColBERT-over-sparse margin at the parent level (99.7% vs 99.4%) suggests that token-level late interaction adds only marginal value when the task reduces to family identification, where learned sparse representations already capture the discriminative lexical patterns necessary for parent-level categorization while maintaining interpretability advantages over dense vectors.

On non-numeric queries, ColBERT and sparse modes achieved perfect accuracy (1.000), while numeric query performance dropped substantially (ColBERT: 44.7%, sparse/dense: 12.5%), confirming ColBERT’s relative robustness to numeric constraints.

4.3.4. Cross-Family Performance Comparisons

Table 11 presents a direct comparison of item-level Acc@1 performance across the best-performing configuration from each model family, revealing substantial performance differences.

The performance hierarchy is clear: BGE-M3-ColBERT (44.8%) outperformed

Table 11: Item-level Acc@1 comparison across neural model families. Values shown for best configuration within each family.

Model Family	Best Configuration	Acc@1
BGE-M3	ColBERT (multi-vector)	0.448 (44.8%)
E5	multilingual-e5-large	0.134 (13.4%)
GTE	gte-large-en-v1.5	0.013 (1.3%)
<i>Single-vector modes within BGE-M3:</i>		
BGE-M3	Sparse	0.125 (12.5%)
BGE-M3	Dense	0.127 (12.7%)

E5-large by 31.4 percentage points and GTE-large by 43.5 percentage points. These substantial differences are meaningful for practical retrieval applications.

4.3.5. Best Neural Models Selected

BGE-M3-ColBERT emerged as the best neural approach (Acc@1=44.8%), achieving 3.3-fold and 34-fold improvements over E5-large and GTE-large respectively. However, even this best neural method retrieved the correct item in less than half of queries.

At the parent level, all neural approaches exceeded 96% Acc@1, confirming that neural models capture broad semantic categories effectively but struggle with fine-grained item-level distinctions.

4.4. Hybrid Fusion Results

Table 12 presents the performance of the best-performing configurations identified through the parameter sweep for each fusion strategy. All optimal configurations converged on $\beta_{\text{exact}} = 0.1$ and $\beta_{\text{overlap}} = 0.0$.

The three-way fusion configurations yielded Acc@1 scores ranging from 0.882 to 0.890, with MRR@10 values between 0.903 and 0.908. The BM25 + ColBERT + TF-IDF combination achieved the highest Acc@1 (0.890) and MRR@10 (0.908).

The five-way fusion produced an Acc@1 of 0.590 and MRR@10 of 0.730, representing substantial decreases of 32.8% and 19.6% respectively compared to the best three-way configuration. However, this strategy achieved the highest Recall@10 (0.963), exceeding the three-way fusions by approximately 1 percentage point. The optimal weight distribution for the five-way fusion allocated 45.0% each to BM25 and TF-IDF, with the three BGE-M3 variants receiving equal 3.3% weights. This weight distribution is itself revealing: the optimizer heavily constrained the neural signals, assigning a collective weight of only 9.9% to all three BGE-M3 variants combined. This distribution suggests that within this highly

Table 12: Performance of hybrid fusion methods with optimized weights. All configurations use $\beta_{\text{exact}} = 0.1$ and $\beta_{\text{overlap}} = 0.0$.

Fusion Strategy	Weight Distribution	Acc@1	R@5	R@10	MRR@10
<i>Three-way fusions (BM25 + BGE-M3 variant + TF-IDF)</i>					
BM25 + ColBERT + TF-IDF	BM25: 0.879 ColBERT: 0.012 TF-IDF: 0.110	0.890	0.941	0.954	0.908
BM25 + Dense + TF-IDF	BM25: 0.879 Dense: 0.012 TF-IDF: 0.110	0.889	0.939	0.953	0.907
BM25 + Sparse + TF-IDF	BM25: 0.879 Sparse: 0.012 TF-IDF: 0.110	0.882	0.940	0.953	0.903
<i>Five-way fusion (BM25 + all BGE-M3 variants + TF-IDF)</i>					
BM25 + Multi-BGE + TF-IDF	BM25: 0.450 ColBERT: 0.033 Dense: 0.033 Sparse: 0.033 TF-IDF: 0.450	0.590	0.928	0.963	0.730

homogeneous domain, dense neural embeddings offer limited discriminative information beyond what lexical models (BM25 and TF-IDF) capture at the granular item level. Furthermore, given the significant drop in Acc@1, even this minor contribution of neural scores appears sufficient to disrupt top-rank precision. Conversely, the elevated Recall@10 of the five-way fusion confirms that neural signals remain valuable for broader retrieval. This behavior is consistent with the parent-level performance observed across all neural configurations, demonstrating that dense embeddings are effective for coarse-grained localization but struggle with fine-grained item discrimination.

4.5. Cross-Encoder Reranking

Cross-encoder reranking was applied to the top-100 candidates from three retrieval systems: BM25 (unigram), the hybrid approach (BM25 + BGE-ColBERT + TF-IDF character n-grams), and BGE-M3-ColBERT. Two reranking strategies were evaluated: *replace*, which uses cross-encoder scores exclusively to rerank candidates, and *blend*, which combines original retrieval scores with cross-encoder scores using weighted interpolation ($\lambda = 0.6$).

Table 13 presents the performance of both reranking strategies for each retrieval system. The blend strategy consistently outperformed the replace strategy across all three systems. For BM25 (unigram), blend achieved Acc@1 of 0.766 compared

Table 13: Performance comparison of cross-encoder reranking strategies. Replace uses CE scores only; Blend combines original retrieval scores with CE scores ($\lambda = 0.6$).

System	Acc@1	Recall@5	Recall@10	MRR@10
<i>BM25 (unigram)</i>				
Baseline	0.869	0.931	0.950	1.000
Replace	0.434	0.841	0.916	0.607
Blend	0.766	0.942	0.966	0.845
<i>BGE-M3-ColBERT</i>				
Baseline	0.448	0.753	0.785	0.580
Replace	0.299	0.664	0.740	0.455
Blend	0.475	0.764	0.786	0.601
<i>Hybrid (BM25 + BGE-ColBERT + TF-IDF)</i>				
Baseline	0.590	0.928	0.963	0.730
Replace	0.430	0.858	0.932	0.608
Blend	0.799	0.985	0.988	0.885

740 to 0.434 for replace, representing an improvement of +0.332. The hybrid system
741 with blend reranking achieved the highest overall performance: 0.799 Acc@1, 0.985
742 Recall@5, 0.988 Recall@10, and 0.885 MRR@10, with improvements of +0.369,
743 +0.127, +0.056, and +0.277 respectively over replace. BGE-M3-ColBERT blend
744 showed improvements of +0.176 in Acc@1 and +0.146 in MRR@10 compared to
745 replace.

746 The blend strategy consistently outperformed replace across all systems,
747 demonstrating that combining original retrieval signals with cross-encoder scores
748 is essential.

749 4.6. Error Analysis

750 To understand the failure modes of each retrieval approach, we conducted a
751 detailed error analysis on 100 randomly sampled failures per method. Errors were
752 classified into four categories: *Numeric Mismatch* (different specifications), *Lexical*
753 *Confusion* (similar terms), *Near-Duplicate Error* (ambiguous entries), and *Other*.

754 4.6.1. Error Type Distribution

755 Table 14 presents the distribution of error types across the three primary re-
756 trieval methods. Both BM25 and BGE-M3 ColBERT showed similar profiles, with
757 numeric mismatches accounting for approximately 98% of failures. The hybrid
758 method showed a distinct profile, with all errors in the “Other” category.

759 Figure 6 visualizes the same distribution and makes the asymmetry between
760 methods immediately apparent: BM25 and BGE-M3-ColBERT share a near-
761 identical failure profile dominated by numeric mismatches, whereas the hybrid
762 method’s failures concentrate entirely in the “Other” category. The hybrid fusion

Table 14: Error type distribution across retrieval methods (% of failed queries, $n = 100$ per method)

Error Type	BM25 Unigram	BGE-M3 ColBERT	Hybrid
Numeric Mismatch	98.1%	97.9%	—
Lexical Confusion	0.5%	0.5%	—
Other	1.3%	1.6%	100.0%

thus appears to resolve the numeric- and lexical-confusion failure modes that affect each component method in isolation, but at the cost of introducing a new failure mode analyzed in the following subsections.

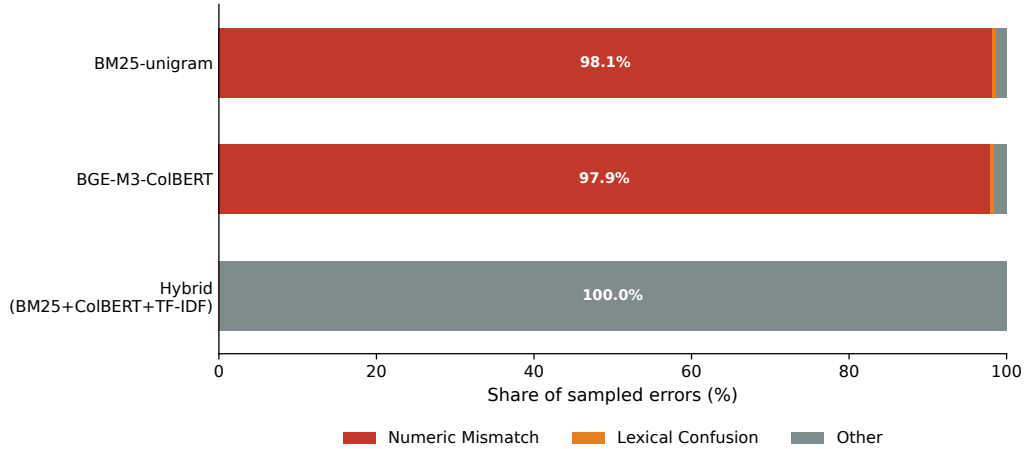


Figure 6: Error type distribution across the three representative retrieval methods ($n = 100$ sampled errors per method). Numeric mismatches dominate both lexical (BM25) and neural (BGE-M3-ColBERT) failures ($\sim 98\%$), revealing a shared blind spot. The hybrid method’s failures concentrate entirely in the “Other” category, a qualitatively different failure mode discussed in Sec. 4.6.

4.6.2. Hierarchical Error Classification

Beyond categorical error types, we analyzed failures according to the hierarchical structure of our construction database, where items are organized under parent categories (e.g., OEB280\$ for 40mm polyethylene conduits). Table 15 shows three distinct failure patterns: (1) retrieving the correct parent category but wrong specific item variant, (2) retrieving an incorrect parent category entirely, and (3) failing to include the correct parent in the top-10 results.

Table 15: Hierarchical error patterns across retrieval methods

Error Pattern	BM25 Unigram	BGE-M3 ColBERT	Hybrid
Right parent, wrong item	10.4%	54.3%	36.0%
Wrong parent	2.6%	0.3%	3.0%
Parent not in top-10	0.1%	0.0%	0.0%

773 The neural method (BGE-M3 ColBERT) demonstrated strong performance at
 774 the parent category level, with 54.3% of errors involving correct parent identifica-
 775 tion but incorrect item selection — more than $5\times$ higher than BM25 (10.4%). This
 776 suggests that dense embeddings effectively capture high-level semantic categories
 777 but struggle with fine-grained distinctions within categories. Conversely, BM25 ex-
 778 hibited higher wrong-parent rates (2.6% vs. 0.3%), indicating occasional failures
 779 at the coarse-grained category level. The hybrid approach showed intermediate
 780 behavior with 36.0% right-parent-wrong-item errors.

781 4.6.3. Representative Error Examples

782 Table 16 presents representative examples. Numeric mismatches occurred
 783 when systems retrieved variants with different specifications (e.g., “3 tubos” vs
 784 “5 tubos”) despite matching other attributes. Lexical confusion involved seman-
 785 tically related but distinct terms. “Other” errors primarily involved mismatched
 786 action verbs or scope modifiers.

787 4.6.4. Method-Specific Error Patterns

788 Analysis of near-duplicate handling revealed stark differences between meth-
 789 ods. BM25 produced 3,276 duplicate documents across 1,602 duplicate groups in
 790 its embedding space, while BGE-M3 ColBERT produced zero duplicates. This dif-
 791 ference stems from BM25’s purely lexical matching, which assigns identical sparse
 792 vectors to textual variants with negligible term frequency differences. The hy-
 793 brid method, incorporating character n-gram features, eliminated these duplicates
 794 while attempting to preserve BM25’s numeric discrimination capabilities.

795 For the hybrid method, the absence of traditional numeric mismatch or lexical
 796 confusion errors suggests that its failures occurred at a different stage of processing.
 797 The 36% right-parent-wrong-item rate indicates that while the method successfully
 798 identified relevant categories, the fusion mechanism between lexical and semantic
 799 signals did not optimally resolve fine-grained distinctions within those categories.

Table 16: Representative error examples by category

Category	Query	Retrieved (Incorrect)	Method
<i>Numeric Mismatch</i>			
OEB290cbebl	Concrete-encased 3 pipes polyethylene 50mm under tracks, any shift, $3 \leq i < 5h$, low volume	Concrete-encased 5 pipes polyethylene 50mm under tracks, any shift, $3 \leq i < 5h$, low volume	BM25
OEB280egecb	Concrete-encased 5 pipes polyethylene 40mm in ballast, any shift, $i < 3h$, low volume	Concrete-encased 5 pipes polyethylene 40mm in ballast, any shift, $3 \leq i < 5h$, low volume	BGE-M3
<i>Lexical Confusion</i>			
OEB250abaec	Supply conduit 4 pipes galv. steel 100mm CMS transitions, exceptional daytime , N/A, any condition	Supply conduit 4 pipes galv. steel 100mm CMS transitions, shift: N/A , maintenance: N/A, any condition	BM25
OEB120aedc	Conduit entry 1–2 pipes in chamber, any shift , no interval needed, any condition	Conduit entry 1–2 pipes in chamber, any exceptional shift , no interval needed, any condition	BGE-M3
<i>Other</i>			
OEB250bbbec	Supply and installation conduit 4 pipes steel 100mm CMS, exceptional night, $i < 3h$, any condition	Supply conduit 4 pipes steel 100mm CMS, exceptional night, $i < 3h$, any condition	BM25
OEB110bda	PVC pipe 110mm chamber drainage, night shift , no interval needed, high volume	PVC pipe 110mm chamber drainage, exceptional night , no interval needed, high volume	BGE-M3

800 4.7. Overall Performance Summary

801 Figure 7 consolidates the central finding visually: the best configuration of
 802 each method family is plotted at both evaluation levels. Lexical retrieval (BM25-
 803 unigram-params) leads at item level by a wide margin, while neural retrieval
 804 (BGE-M3-ColBERT) collapses on the same task — yet recovers to near-perfect
 805 performance at parent level. The dramatic item-to-parent gap exhibited by the
 806 neural family is the visual signature of the paper’s main result: dense embeddings
 807 localize the parametric family well but cannot disambiguate variants within it.
 808 Parent-level metrics were not computed for the Hybrid and Reranking families in
 809 this work and appear as “n/a”.

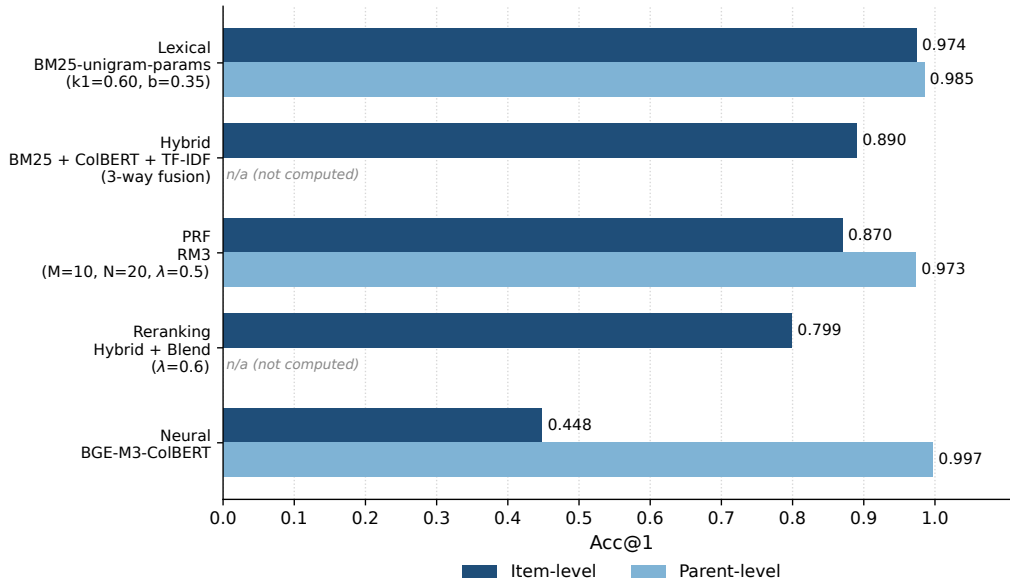


Figure 7: Best Acc@1 per method family at item-level and parent-level on the test set ($n = 16,590$ queries). For each family, the configuration with the highest item-level Acc@1 was selected and both its item- and parent-level scores are plotted. The 52.6-percentage-point item-level gap between Lexical and Neural collapses at parent-level, where Neural reaches 0.997. Parent-level results were not computed for the Hybrid and Reranking families in this work (marked “n/a”); a full parent-level evaluation of these families is deferred to future work.

810 Table 17 presents the top-performing configurations across all evaluated re-
 811 trieval methods, ranked by Acc@1. The table encompasses lexical methods,
 812 pseudo-relevance feedback approaches, neural retrievers, hybrid fusion strategies,
 813 and reranked systems.

Table 17: Top 10 retrieval configurations ranked by Acc@1 on the test set (n=16,590 queries). MRR values provided for reference.

Configuration	Acc@1	MRR
BM25-unigram_params (k1=0.60, b=0.35)	0.974	0.979
BM25 + ColBERT + TF-IDF (3-way fusion)	0.890	0.908
BM25 + Dense + TF-IDF (3-way fusion)	0.889	0.907
BM25 + Sparse + TF-IDF (3-way fusion)	0.882	0.903
RM3 (M=5, N=20, λ =0.5)	0.870	0.898
BM25-unigram (default, k1=0.80, b=0.35)	0.869	0.898
Rocchio (M=5, N=20, β =0.5)	0.857	0.888
BM25-unibigram (k1=0.80, b=0.35)	0.827	0.844
5-way fusion + Blend reranking	0.799	0.885
BM25 + Blend reranking	0.766	0.845
<i>Best neural (for reference):</i>		
BGE-M3-ColBERT	0.448	0.580

814 The evaluation across 100 configurations produced four findings: (1) optimized
815 BM25 achieved the highest performance (Acc@1=97.4%); (2) the best neural ap-
816 proach underperformed by 52.6 percentage points; (3) hybrid fusion nearly dou-
817 bled neural-only performance (89.0% vs 44.8%); and (4) cross-encoder reranking
818 improved results but did not surpass the lexical baseline. These rankings are based
819 on Acc@1, which reflects automated deployment scenarios where only the top result
820 is used. Practitioners operating in semi-automated workflows — where a human
821 reviewer examines the first few candidates — may find Recall@5 or MRR more rel-
822 evant criteria and, under those metrics, hybrid configurations show comparatively
823 stronger performance relative to the lexical baseline, as reflected in Table 17.

824 5. Discussion

825 Our results reaffirm the strength of lexical methods in specialized technical
826 domains. A well-tuned BM25 configuration significantly outperformed all neural
827 retrievers at the item level, achieving approximately 97.4% Acc@1 on a 40,000-
828 item catalog. This aligns with prior findings that optimized lexical models often
829 surpass neural approaches in domain-specific settings with consistent terminology
830 [35], and echoes work showing that term dictionaries improve AEC search precision
831 [36]. For practitioners, the implications are clear: BM25 offers high accuracy
832 with minimal infrastructure and sub-second latency, making it a cost-effective and

scalable solution for catalog retrieval.

Neural retrievers, by contrast, excel at high-level semantic categorization but struggle with fine-grained item matching. At the parent level, models like BGE-M3-ColBERT achieved approximately 99.7% accuracy, rivaling lexical methods; however, at the item level, their performance lagged due to vocabulary mismatch and lack of domain adaptation. General-purpose embeddings underperform on specialized tasks because they handle rare terms poorly [37], and dense retrievers generalize poorly without fine-tuning [38]. Construction catalogs combine technical jargon, codes, and measurements infrequent in general corpora—terms BM25 handles well via inverse document frequency. It is worth noting that this comparison is intentionally asymmetric in one respect: while BM25 was tuned with domain-specific parameter phrases and optimized hyperparameters, neural models were evaluated in zero-shot mode without any domain adaptation. This reflects a realistic deployment scenario where labeled training data is unavailable, and where lexical methods can be configured with modest domain knowledge at low cost. Even if domain adaptation were to narrow this gap, practitioners would still need to weigh the added complexity and computational cost of neural retrievers against the simplicity and transparency of a well-tuned BM25 system.

Hybrid methods improved performance but did not outperform standalone BM25, diverging from general-domain findings where convex combination fusion often outperforms reciprocal rank fusion with minimal training data [39]. In our setting, BM25’s near-optimal accuracy left little headroom for neural signals to contribute meaningfully, suggesting that hybridization may introduce noise in highly technical domains. Learned sparse methods like SPLADE offer a compromise by combining semantic understanding with sparse term expansions [40]; our BGE-sparse variant rivaled ColBERT at parent level, though SPLADE’s complexity remains a deployment barrier.

Cross-encoder reranking showed mixed results. The replace strategy degraded BM25 from 0.869 to 0.434 Acc@1, suggesting that general-domain reranking models struggle to distinguish fine-grained technical differences in parametric catalog entries. The blend strategy ($\lambda = 0.6$) proved more effective, with a hybrid system achieving 0.799 Acc@1. However, BM25 with blend (0.766) still underperformed standalone BM25, suggesting reranking offers limited value when lexical baselines are already strong [41, 42].

Error analysis revealed systematic patterns. Neural retrievers struggled with numeric precision: BERT’s subword tokenization breaks numbers into fragments, impairing magnitude representation [43]—explaining why approximately 98% of neural errors involved numeric mismatches. Many catalog items are near-duplicates differing only in minor attributes, and single-vector models struggled with these fine distinctions whereas ColBERT performed better yet still trailed

lexical methods on exact matches. Construction documents combine structured codes with technical free text, and NLP tools tailored to construction yield high precision by leveraging both cues [44], while specifications encode procedural logic often ignored by generic retrievers [45]. BM25 benefits from transparent matching but remains brittle to synonyms; neural models bridge this gap but introduce opacity.

Prior work focused on classification using rule-based and ML approaches [7, 2, 1, 4, 11], but these studies use small datasets and do not benchmark retrieval across method families. Recent work has noted persistent challenges with construction-specific jargon [2] and domain adaptation for language models [3]. Our study fills this gap by systematically evaluating retrieval on a large-scale catalog, where fine-grained matching reveals effectiveness differences invisible at the category level.

For industry applications, domain-tuned lexical retrieval should be the default. We recommend implementing optimized BM25 with domain-aware preprocessing, conducting targeted error analysis, and exploring neural reranking only when limitations stem from semantic gaps rather than terminology mismatches. Improving data quality often delivers greater returns than model complexity; if neural approaches are pursued, pseudo-label fine-tuning offers a practical path [38].

Information retrieval benchmarks must also account for latency and hardware constraints [46]. BM25 offers sub-100ms latency on standard hardware, while neural retrievers require GPUs with 2–10 $\times$ overhead. Lexical methods accommodate catalog updates without retraining, and interpretability favors BM25’s transparent term matching over opaque neural scores — though learned sparse retrievers offer a middle ground [39]. Retrieval capabilities should be integrated in tools like BIM software to support semi-automated 5D workflows.

This study has several limitations that should be considered when interpreting the results. The dataset used in the experiments covers a single subcategory of the ADIF catalog -specifically, the section dedicated to platform conduit systems (designated as chapter ‘OEB’)—which, while containing nearly 40,000 parametric variants represents a homogeneous domain where all items follow the same structural template and use the same set of descriptive attributes. This homogeneity may favor lexical methods, as BM25 benefits from consistent terminology and predictable parametric vocabulary. At the same time, the near-identical nature of catalog entries positions this homogeneous catalog as a rigorous benchmark for hard-negative retrieval. Validation across other ADIF catalog chapters — such as track infrastructure or electrification systems, which exhibit greater lexical diversity — would strengthen the generalizability of these findings and is a priority for future work. Additional limitations include the zero-shot evaluation of neural models. While this setup reflects realistic deployment constraints where annotated domain-specific data is unavailable, it leaves open the question of how much

domain-adapted fine-tuning could narrow the performance gap with lexical methods. Furthermore, the cross-encoder employed (ms-marco-MiniLM-L-6-v2) was trained exclusively on English web-search data, and its underperformance may reflect the challenges of cross-lingual domain mismatch, rather than an inherent limitation of the reranking paradigm. The poor performance of GTE model (approximately 1% Acc@1) further illustrates that retrieval in technical domains remains an open challenge [47]. Finally, SPLADE was excluded due to limited Spanish language support, and the single-correct-item assumption may oversimplify practical deployment scenarios where multiple catalog entries represent acceptable substitutes.

6. Conclusions and Future Work

We presented a systematic evaluation of retrieval methods for construction parametric catalogs, where small lexical or numeric differences determine success. Across 100 configurations and nearly 40,000 real-world Spanish railway catalog items, the central finding is clear: a well-tuned BM25 configuration incorporating domain-specific parameter phrases achieves 97.4% top-1 accuracy, outperforming the best neural retriever (BGE-M3-ColBERT, 44.8%) by over 52 percentage points and all hybrid configurations by more than 8 points. Neural models, while competitive at the broad category level, consistently fail on fine-grained parametric variants — with nearly all failures occurring on items that differ from the correct answer only in a numeric technical parameter. These results establish lexical retrieval as the best baseline for this class of precision-critical technical search tasks, and demonstrate that neural methods cannot be assumed to generalize effectively to specialized parametric domains without domain adaptation.

For practitioners deploying search over construction catalogs, we recommend starting from a strong lexical baseline with domain-aware tokenization, synonym maps, and unit normalization—favoring simple, well-instrumented baselines before adopting heavier models. Where precision is critical, a compact cross-encoder on top- k candidates improves Acc@1 with modest cost. Exploiting catalog structure by indexing names, types, brands, and standardized attributes separately proves beneficial, as does operationalizing for change: monitoring drift as catalogs evolve, tracking numeric mismatches and near-duplicates, and preferring swappable components.

Several research directions warrant further investigation. Cross-domain validation across other ADIF catalog chapters and external taxonomies would strengthen generalizability [13], while multilingual expansion could address terminology drift and locale-specific conventions [48, 37, 47]. Deployment studies examining informal price-driven queries, efficiency–effectiveness trade-offs [39], and integration

951 with BIM and estimation tools remain necessary, and field validation with con-
952 struction professionals is essential to assess whether ranking improvements yield
953 measurable productivity gains.

954 Technical gaps also persist. Neural encoders need specialized tokenization for
955 numbers, tolerances, and unit conversions. Structured vocabularies and ontologies
956 remain underused despite their utility for disambiguation [49, 50]. Catalog-scale
957 search should optimize both quality and cost/latency [39], and adversarial probes
958 can expose brittleness and prevent silent failures in safety-critical selections [43].

959 Finally, well-measured retrieval is foundational for automation in estimating,
960 compliance checking, and procurement. Our benchmark can accelerate repro-
961 ducible research, while deployment guidance aims to reduce costly mismatches.
962 However, indiscriminate automation may amplify biases or degrade oversight; we
963 recommend phased rollouts with human-in-the-loop review for high-stakes selec-
964 tions.

965 **Appendix A. Supplementary Material**

966 Research data associated with this article will be made available at the follow-
967 ing repository:

- 968 • <https://doi.org/10.5281/zenodo.20277801>
- 969 • <https://doi.org/10.5281/zenodo.20277824>

970 The repository includes:

- 971 • The full parametric construction catalog dataset used in the experiments
- 972 • Query sets, evaluation labels, and retrieval outputs
- 973 • Scripts and configuration files for reproducing the benchmarking results

974 **Appendix B. Technical Background on Retrieval Methods**

975 *Appendix B.1. Information Retrieval Methods*

976 *Appendix B.1.1. Lexical Methods*

977 Traditional information retrieval relies on lexical matching between queries
978 and documents through term-based scoring functions. TF-IDF (Term Frequency-
979 Inverse Document Frequency) weighs terms by their frequency within documents
980 and rarity across the collection [51, 52].

981 BM25 extends TF-IDF with sophisticated length normalization and term fre-
 982 quency saturation [53]:

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d) \cdot (k_1 + 1)}{f(t, d) + k_1 \cdot (1 - b + b \cdot \frac{|d|}{\text{avgdl}})} \quad (\text{B.1})$$

983 where $f(t, d)$ is the frequency of term t in document d , $|d|$ is the document length,
 984 avgdl is the average document length in the collection, k_1 controls term frequency
 985 saturation (typically 0.9–2.0), and b controls length normalization (typically 0.4–
 986 0.8). Recent studies demonstrate that properly tuned BM25 can outperform neural
 987 models in specialized domains, particularly when queries and documents share
 988 consistent terminology [35]. The importance of parameter tuning for BM25 cannot
 989 be overstated — performance can vary significantly across different k_1 and b values
 990 depending on collection characteristics [54].

991 Character n-gram variants of TF-IDF provide robustness to spelling variations
 992 and partial matches, particularly valuable in technical domains with abbrevia-
 993 tions and compound terms [55]. These methods generate overlapping character
 994 sequences (e.g., 3–5 characters), enabling matching even when exact terms differ
 995 slightly.

996 *Appendix B.1.2. Pseudo-Relevance Feedback*

997 Pseudo-relevance feedback (PRF) methods enhance retrieval by assuming top-
 998 ranked documents from an initial search are relevant and using them to reformulate
 999 queries [56].

1000 Relevance Model 3 (RM3) interpolates the original query with a relevance
 1001 model estimated from top-ranked documents [57]:

$$P(t|q') = \lambda P(t|q) + (1 - \lambda) \sum_{d \in R} P(t|d)P(d|q) \quad (\text{B.2})$$

1002 where q' is the expanded query, t is a candidate term, $P(t|q)$ is the term probability
 1003 in the original query, $P(t|d)$ is the term probability in document d , $P(d|q)$ is the
 1004 document relevance score, R is the set of top-ranked pseudo-relevant documents,
 1005 and λ balances original and expansion terms. RM3 consistently improves retrieval
 1006 effectiveness but requires careful tuning of the number of feedback documents and
 1007 terms [58].

1008 Rocchio’s method adjusts query vectors based on centroids of relevant and
 1009 non-relevant documents [56]:

$$\vec{q'} = \alpha \vec{q} + \beta \frac{1}{|R|} \sum_{d \in R} \vec{d} - \gamma \frac{1}{|N|} \sum_{d \in N} \vec{d} \quad (\text{B.3})$$

1010 where $\vec{q}$ and $\vec{q}'$ are the original and modified query vectors, $\vec{d}$ represents document
 1011 vectors, R and N are the sets of relevant and non-relevant documents respectively,
 1012 and α, β, γ control the influence of each component. While originally designed for
 1013 vector space models, Rocchio adapts well to modern retrieval systems [59].

1014 *Appendix B.1.3. Neural Retrieval Methods*

1015 Modern neural approaches transform the retrieval problem into dense vector
 1016 similarity search. Bi-encoder architectures independently encode queries and doc-
 1017 uments into fixed-dimensional representations, enabling efficient similarity compu-
 1018 tation through inner products or cosine similarity [60].

1019 The advent of transformer-based models revolutionized dense retrieval, with
 1020 BERT-based encoders capturing richer semantic relationships [61]. Models like
 1021 Sentence-BERT adapt BERT for efficient sentence-level encoding through siamese
 1022 and triplet networks [31].

1023 Contemporary dense retrieval models include:

- 1024 • **E5 (EmbEddings from bidirectional Encoder rEpresentations):**
 1025 Trained with contrastive learning on large-scale pairs, showing strong zero-
 1026 shot performance [62]
- 1027 • **GTE (General Text Embeddings):** Optimized for diverse retrieval tasks
 1028 through multi-stage training [63]
- 1029 • **BGE (BAAI General Embedding):** Combines contrastive and distilla-
 1030 tion learning for robust representations [32]

1031 Sparse neural methods like SPLADE learn term importance weights while
 1032 maintaining the interpretability of lexical matching [40]. These models use trans-
 1033 former architectures to predict term weights, effectively learning context-aware
 1034 term expansion and reweighting.

1035 Late-interaction models balance effectiveness and efficiency through token-level
 1036 interactions. ColBERT (Contextualized Late Interaction over BERT) encodes
 1037 queries and documents into bags of contextualized embeddings, computing rel-
 1038 evance through the sum of maximum similarities [64]:

$$\text{Score}(q, d) = \sum_{t_q \in q} \max_{t_d \in d} \text{sim}(E_q(t_q), E_d(t_d)) \quad (\text{B.4})$$

1039 Recent multi-representation models, such as BGE-M3, combine dense, sparse,
 1040 and multi-vector strategies within a single framework, adapting to different re-
 1041 trieval scenarios [32]. These unified models show particular promise for multi-
 1042 lingual and domain-specific applications where different representation strategies
 1043 may excel for different query types.

1044 *Appendix B.2. Hybrid and Reranking Approaches*

1045 *Appendix B.2.1. Fusion Methods*

1046 Combining multiple retrieval signals often improves performance by leverag-
1047 ing complementary strengths. Reciprocal Rank Fusion (RRF) merges rankings
1048 without requiring score calibration [65]:

$$\text{RRF}(d) = \sum_{r \in R} \frac{1}{k + \text{rank}_r(d)} \quad (\text{B.5})$$

1049 where $\text{rank}_r(d)$ is document d 's position in ranking r , and k (typically 60) mitigates
1050 the impact of high rankings [66].

1051 Linear combination methods directly interpolate scores after normalization:

$$\text{Score}(d) = \alpha \cdot \text{Score}_{\text{lexical}}(d) + (1 - \alpha) \cdot \text{Score}_{\text{dense}}(d) \quad (\text{B.6})$$

1052 More sophisticated approaches learn combination weights through supervised
1053 learning or employ rank-level fusion considering score distributions [67]. In tech-
1054 nical domains, hybrid methods show particular value when lexical precision and
1055 semantic understanding are both critical [66].

1056 *Appendix B.2.2. Cross-Encoder Reranking*

1057 While bi-encoders encode queries and documents independently for efficiency,
1058 cross-encoders jointly encode query-document pairs, enabling richer interaction
1059 between query and document terms [68]. The cross-attention mechanism in trans-
1060 formers allows fine-grained matching:

$$\text{Score}(q, d) = \text{CrossEncoder}([q; \text{SEP}; d]) \quad (\text{B.7})$$

1061 Cross-encoders consistently outperform bi-encoders in effectiveness but require
1062 computing scores for each query-document pair, making them impractical for large
1063 collections. The standard approach employs cross-encoders to rerank top- k candi-
1064 dates (typically 20–100) from efficient first-stage retrievers [68]. Studies show that
1065 cross-encoder reranking provides consistent improvements across diverse domains
1066 [42].

1067 *Appendix B.3. Evaluation in Technical Domains*

1068 *Appendix B.3.1. Metrics for Precision-Critical Tasks*

1069 Evaluation in technical retrieval requires metrics sensitive to exact matching
1070 and early precision. Mean Reciprocal Rank (MRR) measures the inverse rank of
1071 the first relevant document, particularly suitable for known-item search [69]:

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q} \quad (\text{B.8})$$

1072 where Q is the set of queries and rank_q is the position of the first relevant document
 1073 for query q .

1074 Normalized Discounted Cumulative Gain (nDCG) accounts for graded rele-
 1075 vance and position bias [70]:

$$\text{nDCG@k} = \frac{\text{DCG@k}}{\text{IDCG@k}}, \quad \text{DCG@k} = \sum_{i=1}^k \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)} \quad (\text{B.9})$$

1076 where k is the cutoff rank, rel_i is the relevance score of the document at position
 1077 i , and IDCG@k is the ideal DCG obtained from a perfect ranking.

1078 For technical domains where users expect exact matches, Precision@1 (Suc-
 1079 cess@1) becomes critical, while Recall@k measures the system’s ability to retrieve
 1080 relevant items within reasonable examination depth [71].

1081 *Appendix B.3.2. Statistical Significance Testing*

1082 Proper significance testing is essential when comparing retrieval methods, as
 1083 metrics computed over different query sets can show high variance. Paired statis-
 1084 tical tests account for per-query correlations between systems [72]. The paired
 1085 bootstrap test, sampling queries with replacement, provides robust significance
 1086 estimates without distributional assumptions [34]:

- 1087 1. Resample queries with replacement B times (typically 10,000)
- 1088 2. Compute metric differences for each bootstrap sample
- 1089 3. Estimate p -values from the empirical distribution

1090 Multiple comparison corrections become necessary when testing numerous sys-
 1091 tem pairs. The Bonferroni correction divides the significance threshold by the num-
 1092 ber of comparisons, while less conservative methods like Holm-Bonferroni maintain
 1093 statistical power [73].

1094 *Appendix B.3.3. Domain-Specific Evaluation Challenges*

1095 Technical domains present unique evaluation challenges that standard bench-
 1096 marks may not address. Near-duplicate documents — common in parametric
 1097 catalogs — complicate relevance judgments, as multiple items may satisfy an in-
 1098 formation need [74]. Some evaluation frameworks credit any item from the same
 1099 parametric family (parent-level evaluation), while others require exact matches
 1100 (item-level evaluation) [75].

1101 Numeric parameters in technical documents require special consideration.
 1102 Queries mentioning specific dimensions or quantities should retrieve documents
 1103 with matching values, yet standard text similarity may not capture numeric equiv-
 1104 alence [76]. Prior work in patent retrieval [75], biomedical search [77], and legal
 1105 document retrieval [78] demonstrates the importance of domain-adapted evalua-
 1106 tion protocols that account for these characteristics.

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1112 **CRedit authorship contribution statement**

1113 Cesáreo González-Alvarez: Conceptualization, Methodology, Software, Valida-
1114 tion, Formal analysis, Data curation, Writing – original draft, Visualization.

1115 Laura Fernández-Robles: Conceptualization, Methodology, Software, Vali-
1116 dation, Formal analysis, Data curation, Writing – original draft, Visualization,
1117 Project administration.

1118 Enrique Alegre: Conceptualization, Methodology, Software, Validation, For-
1119 mal analysis, Data curation, Writing – original draft, Visualization, Project ad-
1120 ministration.

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1127 **Data availability statement**

1128 The BC3CAT benchmark is openly available in two archived repositories:
1129 the dataset and BC3/FIEBDC processing pipeline at [https://github.com/cga-](https://github.com/cga-telice/bc3cat-dataset)
1130 [telice/bc3cat-dataset](https://github.com/cga-telice/bc3cat-dataset) (archived on Zenodo, DOI: 10.5281/zenodo.20277801), and
1131 the retrieval and evaluation code at <https://github.com/cga-telice/bc3cat-retrieval>
1132 (archived on Zenodo, DOI: 10.5281/zenodo.20277824). The raw catalog derives
1133 from ADIF's publicly available Base de Precios (<https://bpa.adif.es>).

1134 **Declaration of generative AI and AI-assisted technologies in the** 1135 **manuscript preparation process**

1136 During the preparation of this work the authors used OpenAI ChatGPT and
1137 Anthropic Claude in order to improve language and readability and to assist with

code examples. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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